Project: Setting Up VSFTPD Server and Client on Rocky Linux 9 (Sylvain Simo Kamdem)

This project demonstrates how to install, configure, and verify a fully functional FTP service using vsftpd
(Very Secure FTP Daemon) on Rocky Linux 9, following the best practices from TecMint's RHEL 8 guide.

Objective

Set up a FTP server using vsftpd and connect to it from a client system to upload and download files.
You will configure anonymous and local user access and secure the connection.

Environment

- Server: Rocky Linux 9 (vsftpd server)
- Client: Rocky Linux 9 (FTP client)
- Network: Both systems must have IP connectivity
- Privileges: Root or sudo access

Part 1: Install and Start vsftpd Service

1. Update system packages:
   sudo dnf update -y

2. Install vsftpd:
   sudo dnf install vsftpd -y

3. Enable and start the service:
   sudo systemctl enable vsftpd
   sudo systemctl start vsftpd

4. Verify service status:

sudo systemctl status vsftpd

```
[sims@localhost ~]$ sudo systemctl status vsftpd
● vsftpd.service - Vsftpd ftp daemon
     Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: disabled)
     Active: active (running) since Sat 2025-11-08 15:39:20 CST; 10s ago
    Process: 5379 ExecStart=/usr/sbin/vsftpd /etc/vsftpd/vsftpd.conf (code=exited, status=0/SUCCESS)
   Main PID: 5380 (vsftpd)
      Tasks: 1 (limit: 23124)
     Memory: 740.0K
        CPU: 6ms
     CGroup: /system.slice/vsftpd.service
             └─5380 /usr/sbin/vsftpd /etc/vsftpd/vsftpd.conf
```

Part 2: Configure Firewall for FTP

1. Open FTP service ports in the firewall:

 # firewall-cmd --zone=public --permanent --add-port=21/tcp
 # firewall-cmd --zone=public --permanent --add-port=45073/tcp
 # firewall-cmd --reload

2. Verify open ports:

sudo firewall-cmd --list-all

```
[sims@localhost ~]$ sudo firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh
  ports: 21/tcp 45073/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

Part 3: Configure vsftpd

1. Take the backup of the main FTP configuration file /etc/vsftpd/vsftpd.conf

cp /etc/vsftpd/vsftpd.conf /etc/vsftpd/vsftpd.conf.orig

2. Open and edit the configuration file:

sudo vi /etc/vsftpd/vsftpd.conf

3. Ensure the following parameters are set (uncomment or add them):

anonymous_enable=NO
local_enable=YES

write_enable=YES
local_umask=022
dirmessage_enable=YES
xferlog_enable=YES
connect_from_port_20=YES
xferlog_std_format=YES
listen=NO
listen_ipv6=YES
pam_service_name=vsftpd

5. Configure FTP to allow/deny users access to FTP services based on the user list file **/etc/vsftpd.userlist**.
userlist_enable=YES # allow access to list of usernames from the **userlist_file**
userlist_file=/etc/vsftpd.userlist # stores usernames.
userlist_deny=NO

6. Add the following lines in your vsftpd.conf configuration file to restrict user access to their home directory:
chroot_local_user=YES #means local users will be placed in a chroot jail, their home directory after login by default settings.
user_sub_token=$USER
local_root=/home/$USER/ftp

Part 4: SELinux Configuration (if enforced)

If SELinux is enabled, apply the correct context to allow FTP read/write operations:
sudo semanage boolean -m ftpd_full_access --on

sudo setsebool -P ftp_home_dir on
sudo setsebool -P ftpd_full_access on

Restart the vsftpd service:
sudo systemctl restart vsftpd

Part 5: Create FTP User

1. Add a new local user for FTP:
sudo adduser -m -c "Sims FTP user" -s /bin/bash ftpuser
sudo passwd ftpuser

2. Add the user **ftpuser** to the file **/etc/vsftpd.userlist**
 # echo "ftpuser" | tee -a /etc/vsftpd.userlist
 # cat /etc/vsftpd.userlist

3. Create the alternative local root directory for the user and set the appropriate permissions on this directory:

sudo mkdir -p /home/ftpuser/ftp
sudo chown nobody:nobody /home/ftpuser/ftp
sudo chmod a-w /home/ftpuser/ftp

3. Create directory inside the local root location, where user's will keep his/her files:

mkdir /home/ftpuser/ftp/files

sudo chown ftpuser:ftpuser /home/ftpuser/ftp/files
sudo chmod 0700 /home/ftpuser/ftp/files/

Part 6: FTP Client Configuration

1. Install FTP client:

sudo dnf install ftp -y

2. Connect to the FTP server:

ftp ftpuser@ip_address

```
[sims@localhost ~]$ ftp ftpuser@192.168.1.236
ftp: ftpuser@192.168.1.236: Name or service not known
ftp> _
```

4. Upload and download test files:

put testfile.txt
get testfile.txt

Part 7: Verify FTP Logs

Check log files for activity and troubleshooting:

sudo tail -f /var/log/vsftpd.log

Part 8: Security Hardening

- Disable anonymous access (ensure anonymous_enable=NO).
- Use FTPS (SSL/TLS) for encryption.
- Limit users with /etc/vsftpd/user_list.

- Configure passive ports if using NAT/firewalls:
  pasv_min_port=40000
  pasv_max_port=50000

Verification Checklist

☑ vsftpd service is running
☑ FTP port 21 is open and accessible
☑ Client can connect and authenticate
☑ Upload and download work successfully
☑ Logs show transfer activity

Deliverables

1. Screenshot of successful FTP login from client.
2. Output of 'systemctl status vsftpd' on server.
3. Screenshot of uploaded file in /home/ftpuser/ftp_upload.